Project3_Netflix_Analysis.ipynb

April 17, 2026

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

print("  All libraries imported successfully!")
```

```
Matplotlib is building the font cache; this may take a moment.

  All libraries imported successfully!
```

[2]:
```
df = pd.read_csv('https://raw.githubusercontent.com/prasertcbs/basic-dataset/
↪master/netflix_titles.csv')
df.head()
```

[2]:
```
    show_id     type                                    title  \
0  81145628    Movie  Norm of the North: King Sized Adventure
1  80117401    Movie               Jandino: Whatever it Takes
2  70234439  TV Show                       Transformers Prime
3  80058654  TV Show         Transformers: Robots in Disguise
4  80125979    Movie                             #realityhigh

                   director  \
0  Richard Finn, Tim Maltby
1                       NaN
2                       NaN
3                       NaN
4          Fernando Lebrija

                                              cast  \
0  Alan Marriott, Andrew Toth, Brian Dobson, Cole…
1                                 Jandino Asporaat
2  Peter Cullen, Sumalee Montano, Frank Welker, J…
3  Will Friedle, Darren Criss, Constance Zimmer, …
4  Nesta Cooper, Kate Walsh, John Michael Higgins…
```

```
                                 country         date_added  release_year  \
0  United States, India, South Korea, China  September 9, 2019          2019
1                            United Kingdom  September 9, 2016          2016
2                             United States  September 8, 2018          2013
3                             United States  September 8, 2018          2016
4                             United States  September 8, 2017          2017

     rating  duration                            listed_in  \
0     TV-PG    90 min  Children & Family Movies, Comedies
1     TV-MA    94 min                     Stand-Up Comedy
2  TV-Y7-FV  1 Season                            Kids' TV
3     TV-Y7  1 Season                            Kids' TV
4     TV-14    99 min                            Comedies

                                         description
0  Before planning an awesome wedding for his gra…
1  Jandino Asporaat riffs on the challenges of ra…
2  With the help of three human allies, the Autob…
3  When a prison ship crash unleashes hundreds of…
4  When nerdy high schooler Dani finally attracts…
```

[3]:

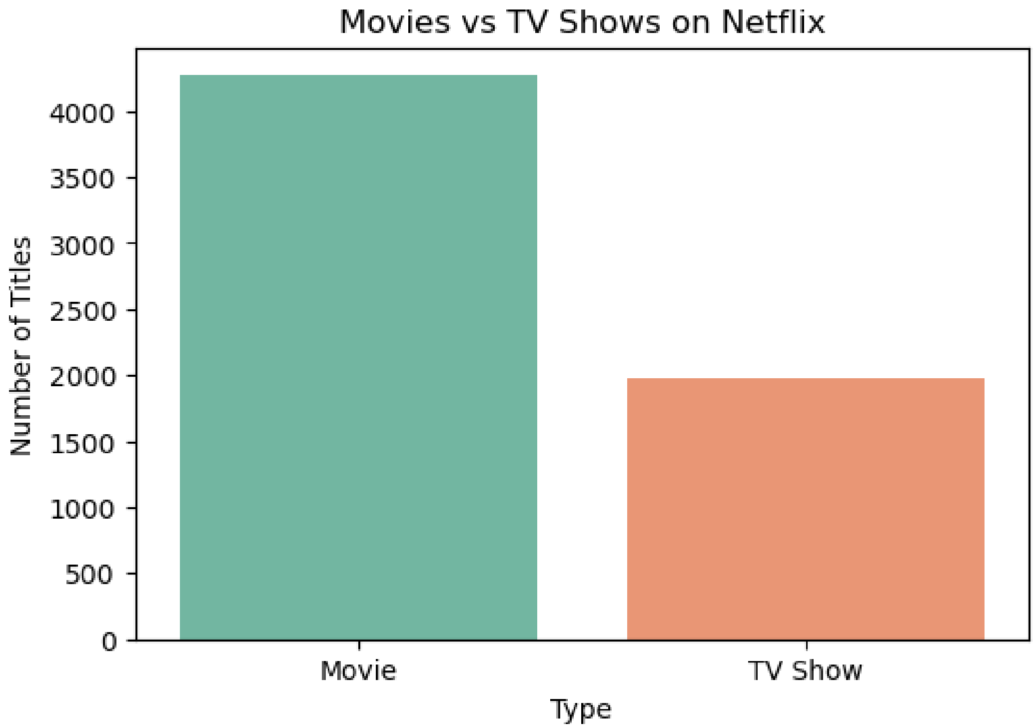
```python
print("Shape of data (rows, columns):", df.shape)
print("\nColumn names:\n", df.columns.tolist())
print("\nData types:\n", df.dtypes)
print("\nMissing values in each column:\n", df.isnull().sum())
df['director'] = df['director'].fillna('Unknown')
df['cast'] = df['cast'].fillna('Unknown')
df['country'] = df['country'].fillna('Unknown')
df['rating'] = df['rating'].fillna('Unknown')
df['duration'] = df['duration'].fillna('Unknown')
print("\n  Missing values fixed with 'Unknown' where needed!")
```

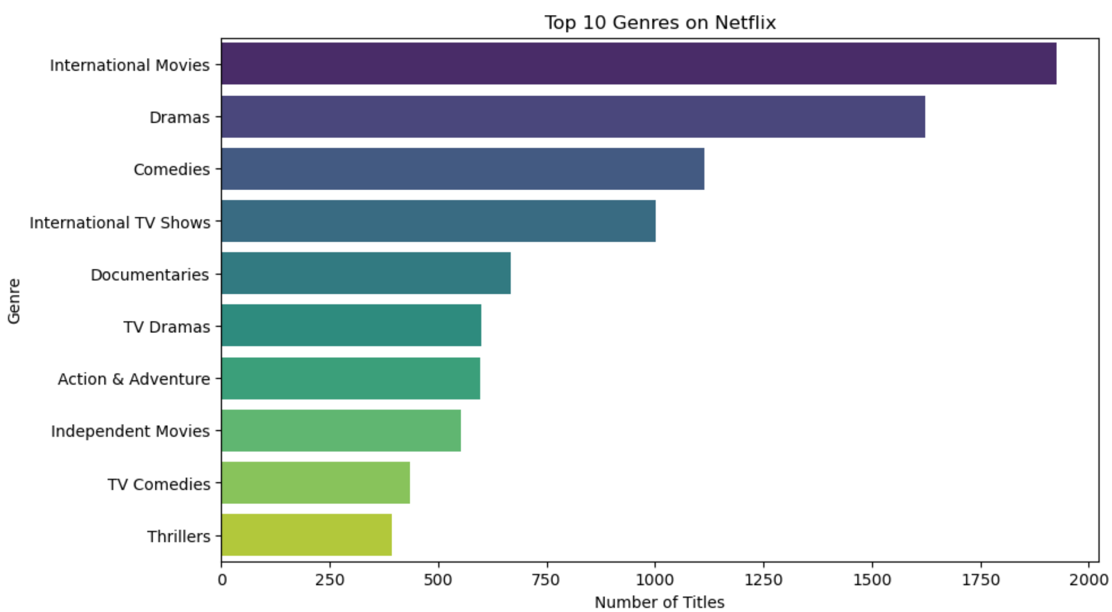
```
Shape of data (rows, columns): (6234, 12)

Column names:
 ['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added',
'release_year', 'rating', 'duration', 'listed_in', 'description']

Data types:
 show_id          int64
type            object
title           object
director        object
cast            object
country         object
date_added      object
release_year     int64
```

```
rating          object
duration        object
listed_in       object
description     object
dtype: object

Missing values in each column:
 show_id           0
type              0
title             0
director       1969
cast            570
country         476
date_added       11
release_year      0
rating           10
duration          0
listed_in         0
description       0
dtype: int64

 Missing values fixed with 'Unknown' where needed!
```

[4]:

```python
import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(6,4))
sns.countplot(x='type', data=df, palette='Set2')
plt.title('Movies vs TV Shows on Netflix')
plt.xlabel('Type')
plt.ylabel('Number of Titles')
plt.show()
plt.figure(figsize=(10,6))
genres = df['listed_in'].str.split(', ', expand=True).stack().value_counts().
↪head(10)
sns.barplot(y=genres.index, x=genres.values, palette='viridis')
plt.title('Top 10 Genres on Netflix')
plt.xlabel('Number of Titles')
plt.ylabel('Genre')
plt.show()
print("  Two beautiful plots created successfully!")
```

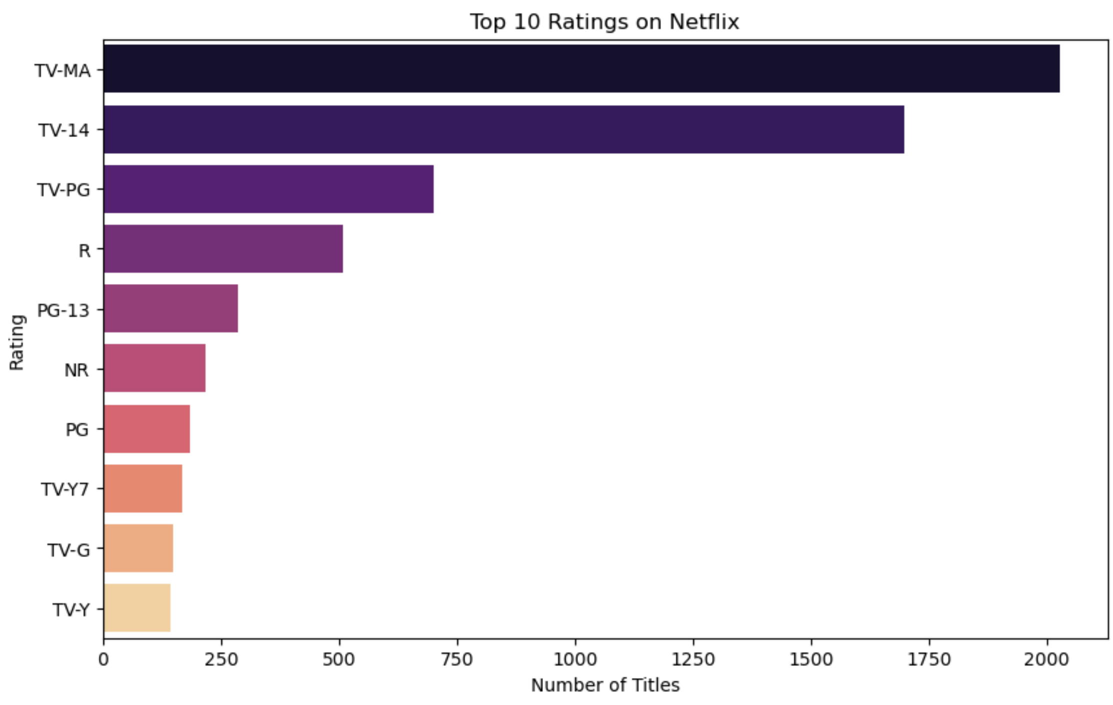


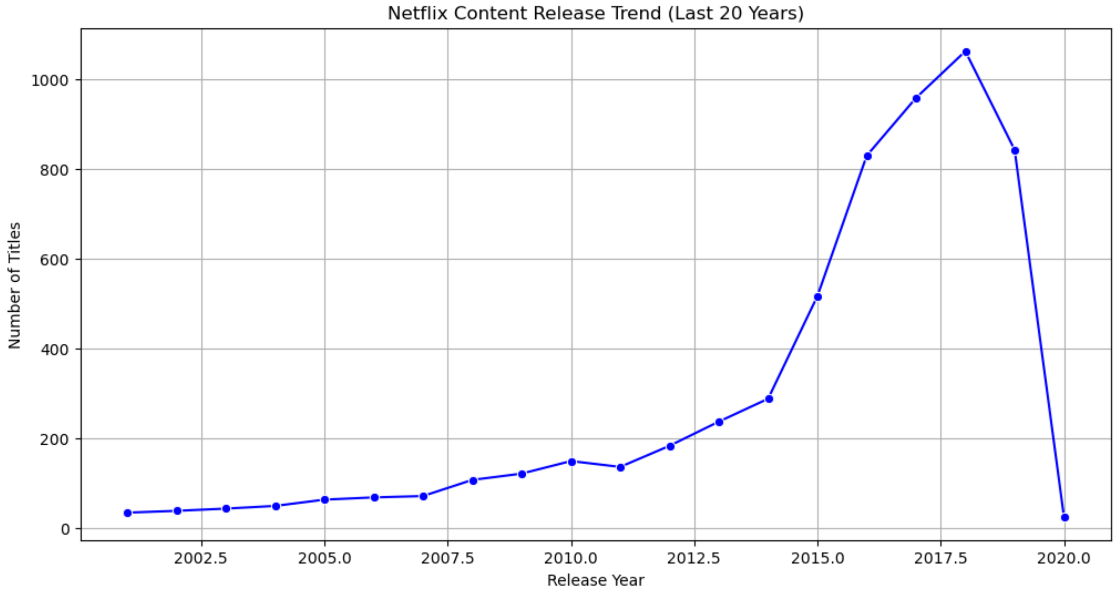



```
Two beautiful plots created successfully!
```

[5]:

```
import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(10,6))
rating_counts = df['rating'].value_counts().head(10)
sns.barplot(y=rating_counts.index, x=rating_counts.values, palette='magma')
plt.title('Top 10 Ratings on Netflix')
plt.xlabel('Number of Titles')
plt.ylabel('Rating')
plt.show()
plt.figure(figsize=(12,6))
year_counts = df['release_year'].value_counts().sort_index().tail(20)  # last␣
↪20 years
sns.lineplot(x=year_counts.index, y=year_counts.values, marker='o',␣
↪color='blue')
plt.title('Netflix Content Release Trend (Last 20 Years)')
plt.xlabel('Release Year')
plt.ylabel('Number of Titles')
plt.grid(True)
plt.show()

print("  Two more beautiful plots created successfully!")
```



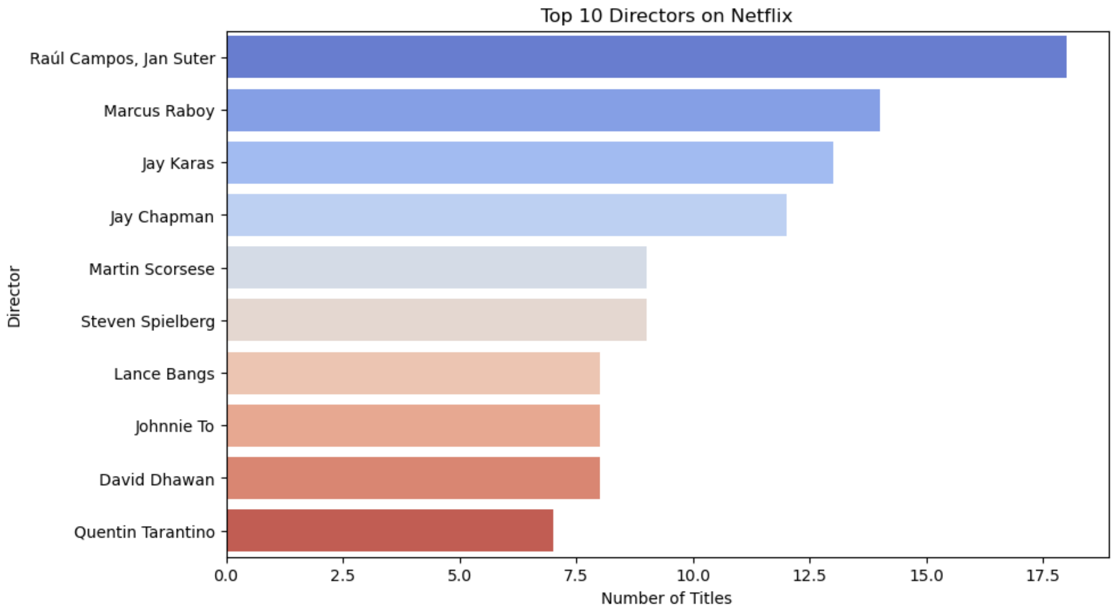


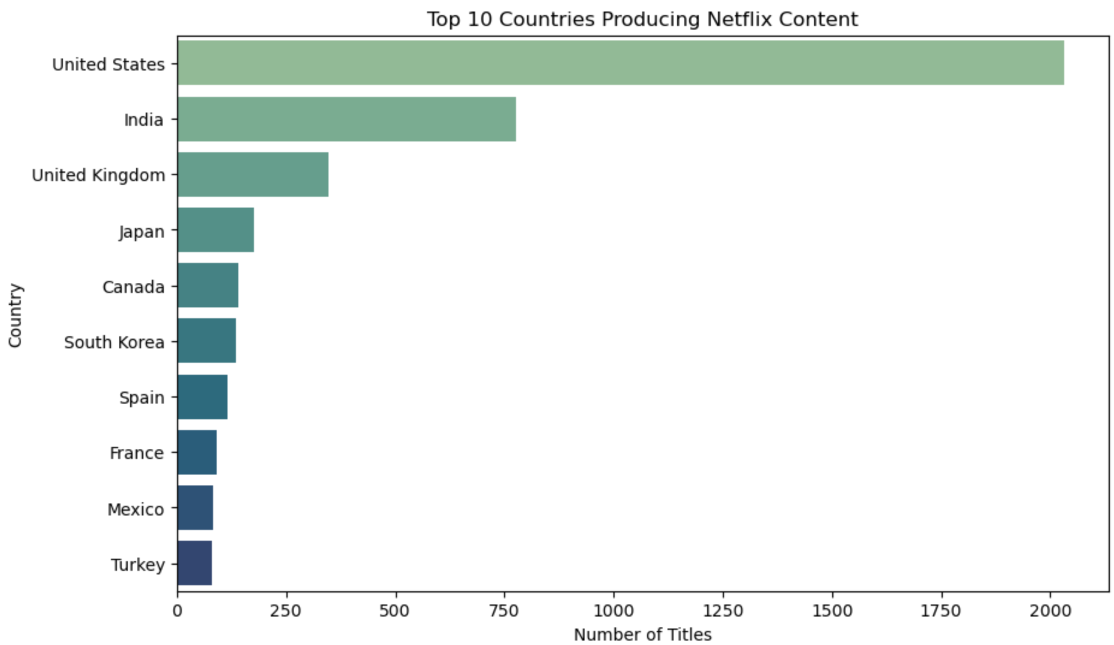

```
Two more beautiful plots created successfully!
```

[6]:

```
import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(10,6))
top_directors = df[df['director'] != 'Unknown']['director'].value_counts().
  ↪head(10)
sns.barplot(y=top_directors.index, x=top_directors.values, palette='coolwarm')
plt.title('Top 10 Directors on Netflix')
plt.xlabel('Number of Titles')
plt.ylabel('Director')
plt.show()
plt.figure(figsize=(10,6))
top_countries = df[df['country'] != 'Unknown']['country'].value_counts().
  ↪head(10)
sns.barplot(y=top_countries.index, x=top_countries.values, palette='crest')
plt.title('Top 10 Countries Producing Netflix Content')
plt.xlabel('Number of Titles')
plt.ylabel('Country')
plt.show()
print("Pattern Detection Insights:")
print("1. Most common content type      :", df['type'].value_counts().index[0])
print("2. Most popular rating           :", df['rating'].value_counts().
  ↪index[0])
print("3. Most common genre             :", df['listed_in'].str.split(', ').
  ↪explode().value_counts().index[0])
```

```
print("4. Total titles from USA          :", (df['country'] == 'United States').
 ↪sum())
print("5. Average release year           :", round(df['release_year'].mean(), 0))

print("\n Two more plots + insights created successfully!")
```

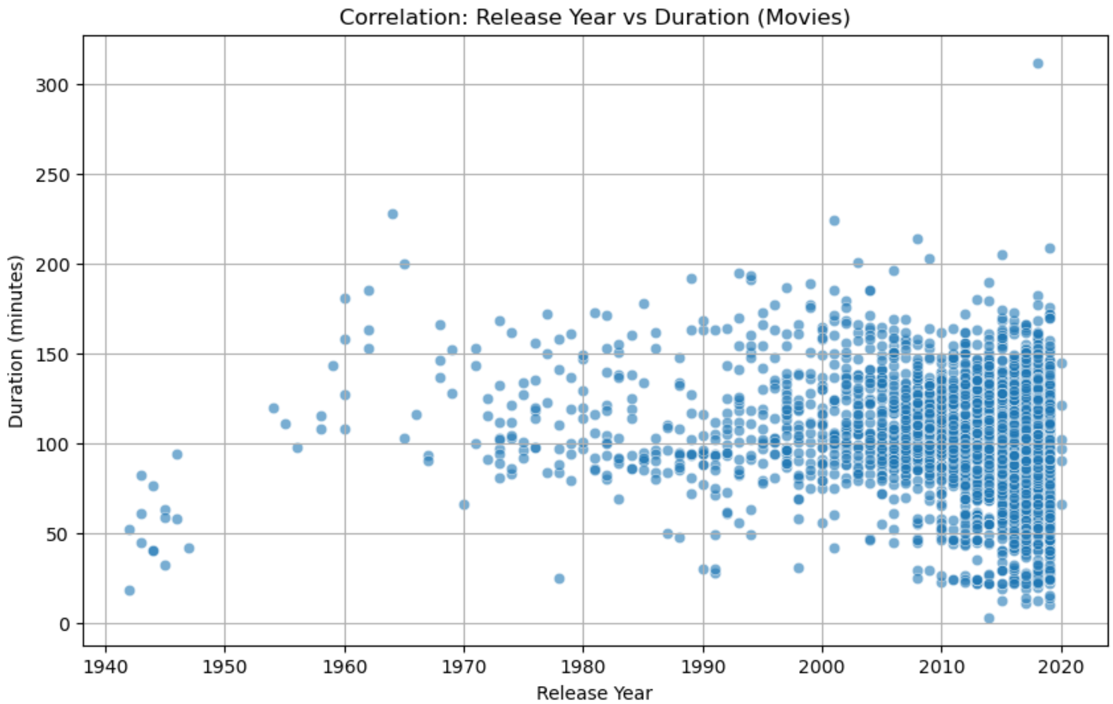




```
Pattern Detection Insights:
1. Most common content type      : Movie
2. Most popular rating           : TV-MA
3. Most common genre             : International Movies
4. Total titles from USA         : 2032
5. Average release year          : 2013.0

 Two more plots + insights created successfully!
```

[12]:

```python
import matplotlib.pyplot as plt
import seaborn as sns
df['duration_num'] = df['duration'].str.extract('(\d+)').astype(float)
plt.figure(figsize=(10,6))
movies = df[df['type'] == 'Movie']
sns.scatterplot(x='release_year', y='duration_num', data=movies, alpha=0.6)
plt.title('Correlation: Release Year vs Duration (Movies)')
plt.xlabel('Release Year')
plt.ylabel('Duration (minutes)')
plt.grid(True)
plt.show()
numeric_df = df[['release_year', 'duration_num']].dropna()
corr_matrix = numeric_df.corr()

print("  Correlation Analysis:")
print("Correlation between Release Year and Duration:", round(corr_matrix.
 ↪loc['release_year', 'duration_num'], 3))
print("\n" + "="*50)
print("  PROJECT 3 COMPLETE - NETFLIX USER BEHAVIOR ANALYSIS")
print("="*50)
print("Key Insights:")
print("• More Movies than TV Shows on Netflix")
print("• Top Genres: Dramas, Comedies, Documentaries")
print("• Most Common Rating: TV-MA")
print("• Content is growing fast after 2015")
print("• USA produces the most content")
print("• Newer movies tend to be slightly shorter correlation shown above")
```




```
 Correlation Analysis:
Correlation between Release Year and Duration: -0.221

==================================================
 PROJECT 3 COMPLETE - NETFLIX USER BEHAVIOR ANALYSIS
==================================================
Key Insights:
• More Movies than TV Shows on Netflix
• Top Genres: Dramas, Comedies, Documentaries
• Most Common Rating: TV-MA
• Content is growing fast after 2015
• USA produces the most content
• Newer movies tend to be slightly shorter correlation shown above
```

[ ]: